Chloe Kim

Resume – Academic Emphasis

Cranbrook Kingswood Upper School, 39221 Woodward Ave. Bloomfield Hills, MI

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Academic Interests & Selected Outputs

Focus: AI × Human Decision-making under Uncertainty & Fairness

- Cognition & intelligence — perception, reasoning, risk
- Applied domains: social applications of AI, AI insurance, legal networks, healthcare triage

Keywords: Computational cognitive science; reinforcement learning & Bayesian inference; NLP; causal inference

Selected Outputs

- *Professional Academic Presentations (3):* EURO 2025; Helmholtz AI 2025 (*Spotlight*)
  — Human *vs.* model-based reasoning; Recognition & uncertainty, fairness; AI applications
- *Research Papers (2):* PLOS One; Cognitive Systems Research (submitted)
  — Emergent architecture of law; Intuition *vs.* semantic reasoning by humans and AI
- *Patents (2):* AI insurance framework; Influence scoring of judicial decisions (Sole Inventor)
- *Software Reg. (4 works):* Network dynamics, power-law & inequality study (Sole Developer)

Education

Cranbrook Kingswood Upper School *High school student* — *Aug 2022–Jun 2026 (expected)*
Bloomfield Hills, MI

Harvard University *Undergraduate-level studies* — *July 2024–Aug 2024*
Cambridge, MA

Molecular Biology (BIOS-S1a) (A-) and Historical Linguistics (LING-120) (A)
Studies at Harvard Summer School (for credit)

南京时代双语学校 (Nanjing Times College) *Middle school student* — *Apr 2020–Jun 2022*
南京(Nanjing), China

Kyunghee School *Elementary school student* — *Mar 2014–Feb 2020*
Seoul, Republic of Korea

Selected Research & Intellectual Property

Patents *Sole Inventor*

"System and method for evaluating AI output consistency and providing dynamic insurance"
#10-2025-0104581, patent pending (Korea)

- Quantifies AI risks (factuality, logical/semantic consistency, bias) into actuarial metrics.
- Designs dynamic premium pricing & claim validation linked to AI performance.
- *AI Insurance* – recompensation for AI hallucinations

"Systems and Methods for Temporal Influence Scoring of Judicial Decisions"
#10-2025-0166979, patent pending (Korea)

- Builds a dated, directed legal-citation index; computes influence measures (adoption/impact).
- Improves retrieval via structure-weighted rankings and rolling-window alerts that surface pivotal precedents and enable monitoring.

Software Registrations *Sole Developer*

"Interpreting Social Intent and Behavior: Temporal Dynamics of Network Heatmaps" *(#035490)*

- Analyzes temporal networks, creating centrality- & time-based maps of social relations.

"Analysis of Social Intentionality: Visualization of Case-law Networks" *(#035307)*

- Processes legal datasets, computing centrality & visualizing case-law structures.

"Simulation of Behavioral Relationships in Power-law Dynamics using AI" *(#035489)*

- Models behavior relations under power-law dynamics with AI sampling & visualization.

"Inequalities in Human Behavior: A Graph-based and AI Interpretation" *(#034662)*

- Analyzes behavioral inequality using AI/graphs, quantifying disparities via references.

Research Presented & Papers *First author & Presenter*

"Dynamic Network Analysis of Legal Citation Networks (especially medical)"
34th European Conference on Operational Research (EURO 2025), Leeds, UK

- Analyzed 86,000+ Supreme Court cases using network and power-law analysis, showing concentrated reliance on landmark precedents.
- Illustrates how structural citation patterns shape legal & medical decision-making.

"Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)"
PLOS One: under review (Oct 2025)

"An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI"
Helmholtz AI Conference 2025, Karlsruhe, Germany

- Compared 40+ multilingual participants with GPT-4 on sound-to-shape tasks, revealing humans' intuitive mappings *vs.* AI's semantic reasoning.
- Findings support dual-mode cognition, highlighting differences in human and machine decision-making.

"Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination"
Cognitive Systems Research: under review (Dec 2025)

"Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction" *(Spotlight)*
Helmholtz AI Conference 2025, Karlsruhe, Germany

- Trained on 500+ patient cases with fused imaging (Vision Transformer) and clinical features (XGBoost), achieving 92.7% accuracy and AUROC 0.90.
- Results demonstrate scalable, interpretable AI for airway management, highlighting contrasts between human judgment and standardized decision-making.

Work Experience

Lawfirm Hoam *Research Assistant* *Summer 2024–*
Seoul, Rep. of Korea & Remote

- Conduct data (legal) analytics and case research using AI and quant. network analysis.
- Presented results at an academic conference.

Quilt.ai *Research Intern* *Jul 2024–Dec 2024*
Boston, MA & Remote

- Evaluated and researched AI narrative generation for fashion ads that support human perception and cognition.
- Ran market surveys; produced insight memos used by the design/product teams.

Media art Yeseung Lee's team *Assistant Editor/Supporter* *Dec 2024–*
Seoul, Rep. of Korea & Remote

- Supported a project on art and human cognition.
- Edited artist statements and exhibition texts (Singapore, Seoul Dongdaemun Design Plaza (DDP), National Museum of Modern Art (MMCA) exhibitions).

Elroi AI *Research Intern* *May 2025–*
Troy, MI & Remote

- Prototyped and studied modules for generation & HAI workflows; conducted user tests.
- Delivered a technical brief.

Herring Insight *Volunteer Analyst* *July 2024–*
Newton, MA

- Analyzed cultural-difference on how to support deprived people's education.

SAT Mathematics Tutoring *Tutor* *June 2025–September 2025*
NYC, NY & Remote

- Tutored high school students in SAT Math preparation and strategies.

Computer Science Projects *Independent researcher* *Jun 2022–*
Bloomfield Hills, MI & remote

- Built coding projects including BMI Calculator, Quiz Game, Server–Client Trading Apps, and AI&ML-based game simulations. (9th – 12th grade)

Leadership & Community Service

Varsity Rowing Team *Coxswain*

- Coach/manage boats (steering/making calls & corrections), mentor novice rowers, delegate roles, manage race day logistics & encourage team support

Linguistics Club *Founder, Co-President*

- Founded Linguistics Club to promote different branches of linguistics; raise awareness of diverse languages; NACLO prep; AI+ Linguistics research

Crane Clarion (School newspaper) *Assoc. Arts Editor, Graphic Designer, Staff Writer*

- Led artists & writers; mentored junior writers; established journalistic protocols; publicized arts & school spirit; wrote cultural sens. guidelines

Ethics Bowl Club *President/Founder*

- Founded club; recruited, hosted state-wide practice events, organized spaces for ethical thinking & dialogue in & out of school & community

LITE *(Leaders in the Environment)* *Food Board Captain*

- Liaison to school meal providers; promoted inclusive meals; systemized negotiation process btw. student body, faculty, and providers; org 10+ events

Cranbrook Kingswood Chamber Orchestra, Thornlea Quartet *Violinist; Organizer*

- Violinist in school orchestra; performed at school/community; organized fundraising concerts for Brilliant Detroit; mentored peers

Tutoring *Comm. Service – Volunteer tutor*

- Independent & Volunteer – Local & Overseas; hybrid—in-person/online via Zoom/Teams)
- Tutored 5-7 under-resourced students: English (Chinese & Korean) + SAT Math (US); ran Zoom lessons; built practice sets; measurable score gains

Gold Key *Sophomore Lead*

"Ergasterion" – Theater & Drama club *Stage Crew & Member*

Student Council *Member*

Awards and Honors

National Latin Exam Cum Laude and 2x Magna Cum Laude – *G9, 10, 11; National*
Ethics Bowl Central Division 4th place – *G10, 11; State*
3rd in State for National Economics Challenge – *G10; State*
National Merit semifinalist – *National*
District MSBOA 1st Division (solo) & 1st/2nd Division (quartet) – *G9, 10, 11; State*
NEC – National Economics Challenge; 3rd (D. Ricardo) – *G10; State*
Strickland Writing Award – *G9; School*
Dean's List & Honors – *G9, 10, 11; School*
AP Scholar with Distinction Award – *G11; School*

Skills

AI and Data Analytics

- Statistics and Data Analytics
- Network and Complex System Analysis

Technical Computer Science: *Java, Python, Julia* – Applications to Semantics and Philosophy

Additional

Languages
English (*native*); Korean (*native/bilingual*); Chinese (Mandarin, *native*); French (*intermediate*); Japanese (*intermediate*); Latin (*NLE Latin III*)

Training (selected external coursework)
MIT OCW (Python/Julia, abstraction, object-oriented programming, recursion); FastCampus ML (TensorFlow, Docker, LLM agents); Human Asia (*"Human Rights"* for North Korea)

Citizenship: United States

Abstracts

Patents

- System and method for evaluating AI output consistency and providing dynamic insurance *(#10-2025-0104581, Pending)*

Abstract translated from the filed application

- *Abstract*

This invention relates to novel systems and methods that address the reliability issues of outputs generated by artificial intelligence such as large language models (LLMs) and that provide insurance coverage for the associated risks.

The invention organically integrates an "AI Output Consistency Evaluation Module," which assesses the reliability of AI outputs across multiple dimensions, with an "AI Output Insurance System," which operates insurance products based on the evaluated risks. The evaluation module comprehensively verifies factors such as factual accuracy, logical and semantic consistency, and bias to generate a quantitative composite risk profile.

The insurance system directly ingests this profile and dynamically calculates and adjusts premiums and coverage terms in line with the model's performance; in the event of an incident, it verifies the validity of claims based on pre-recorded, objective performance data.

Through this approach, the invention converts the unpredictable technological risks of AI into manageable insurance liabilities and promotes the responsible adoption and use of AI by enterprises.

- Systems and Methods for Temporal Influence Scoring of Judicial Decisions
  (#10-2025-0166979, Pending)

- *Abstract*

  A computer-implemented system generates time-scoped influence signals for legal search and monitoring by dynamically analyzing citation networks of judicial decisions. It ingests a large corpus with citation metadata, normalizes references, and builds a dated, directed citation index. On this index, the system computes two complementary measures for each decision: an adoption index (in-degree–based reliance by later cases) and an impact index (recursively computed from citation relations). Scores are evaluated in rolling time windows and can be linked to subject labels to produce interpretable timelines and dashboards. Outputs include structure-weighted rankings, early surfacing of pivotal precedents, and alerts when influence moves across topics or time bands, with rationales that show contributing sources and time weights.

Research Presented & Papers

- Dynamic Network Analysis of Legal Citation Networks (esp. medical) in South Korea (1951–2023)

34th European Conference on Operational Research (EURO)

- *Abstract*

We analyze approximately 86,000 legal cases from South Korea (1951–2023) by constructing (complex) citation networks, modeling each case as a node and citation relationships as edges. Through centrality/hub analysis, we identify key hub cases. We extract and analyze a focused subnetwork comprising medical-related cases.

Segmenting the network along presidential cycles reveals that the distribution of medical hub cases changes over time. Cases related to criminal liability in medical accidents were prominent hubs from the late 1990s to early 2000s, whereas subsequent administrations saw civil rulings on patients' rights and institutional responsibilities become new hubs. This analysis illustrates structural reorganization within the network.

From an optimization perspective, we apply the minimum dominating set (MDS) approach, supplemented by AI-assisted heuristics, to extract a minimal influential subset within the medical subnetwork. Complexity analysis shows power-law degree distributions with the exponent approximately 2.1, indicating scale-free characteristics. An AI-assisted optimization, balancing sparsification and policy relevance, facilitates a more targeted identification of legally and policy-relevant cases.

We suggest quantitative investigations into the structural features and dynamic evolution of legal citation networks in the medical domain. These provide practical implications for legal and policy decision-making in healthcare governance.

- *How I obtained the opportunity*
  I secured the opportunity through direct outreach and an interview with Attorney Seongwoong Kim (Lawfirm Hoam); I collaborated under his mentorship. My research was accepted and presented at the conference.

- *Primary author (if not me)*
  I am the first author and presenter.

- *Scholarly Guidance and Oversight*
  Seongwoong Kim – Mentor, Co-author

- Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)

PLOS One *(an academic journal)*
Manuscript Submitted for Review

- *Abstract*

The evolution of a nation's legal system can be read as a data-driven and AI narrative of its socio-political development. Conceptualizing law as a complex adaptive system, we conduct the first large-scale dynamic network analysis of the Republic of Korea's Supreme Court jurisprudence, mapping 86,000+ decisions from 1951–2023 into a comprehensive citation network. The network is not random but scale-free, with a power-law distribution ($\alpha = 3.29$) consistent with preferential attachment, and exhibits substantial inequality (Gini coefficient of 0.604), whereby a small number of landmark cases operate as influential hubs. Crucially, we distinguish two modes of precedent influence: popularity (in-degree) versus authority (centrality). Highly cited cases are not always the most authoritative; popular precedents often serve as routine tools, whereas high authority cases constitute the backbone of legal reasoning. A focused examination of the medical-law sub-network reveals an even more hierarchical structure ($\alpha = 2.14$), indicating that knowledge specialization amplifies structural inequality. A temporal segmentation by presidential administrations uncovers a marked semantic shift in influential medical jurisprudence: from state-centric criminal enforcement in earlier periods to citizen-centric civil and administrative law emphasizing patient rights and professional accountability in later democratic eras. Viewed through a cognitive lens, the citation network functions as a collective mind: judicial behavior, which is driven by bounded cognition and heuristic search, produces a brain-like architecture that stores institutional memory and channels inference through authoritative hubs. This dual structural-semantic account offers a behavioral and cognitive framework for understanding how legal doctrine co-evolves with societal change, with implications for legal theory, judicial behavior, policy design, and the development of artificial intelligence in law.

- *How I obtained the opportunity*
  I secured the opportunity through direct outreach and an interview with Attorney Seongwoong Kim; I collaborated under his mentorship.

- *Primary author (if not me)*
  I am the first and co-corresponding author.

- *Scholarly Guidance and Oversight*
  Seongwoong Kim – Mentor, Collaborator, Co-author

- An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI

Helmholtz AI Conference 2025

- *Abstract*

We experimentally analyze how association stimuli are processed through distinct cognitive mechanisms by humans and AI (large language models, LLMs) in visual generation tasks based on linguistic association. As stimuli, we employ nonsymbolic sound words (non-symbolic sounds), which represent meaningless but structured sound sequences. This study focuses on the contrast between human intuition and LLMs' learned semantic reasoning. Humans tend to visualize 'sound-associated words' without prior learning, relying on immediate sensory and affective cues to produce intuitive forms—curves or pointed shapes. In contrast, LLMs such as GPT map these words onto shapes based on large-scale text-driven knowledge, using semantic structures and statistical patterns.

To verify this, we conduct experiments with both a diverse cross-linguistic and cross-cultural human participant group (n ≃ 40; including Korean, Chinese, English, French, and Hindi speakers) and an LLM (GPT-4). Shape categorization classifies the imagined (in the case of humans) and generated images (in the case of the LLM) into basic geometric forms such as circles, triangles, and rectangles. Results show that humans rely on pre-existing intuitive mappings between sound and shape, performing direct sensory-to-visual transfers without explicit learning. By contrast, LLM outputs (expression of shapes) are generated through prior semantic knowledge learned from language corpora, reflecting a meaning-driven reasoning process.

We identify a fundamental gap between human intuition-based learning and LLMs' meaning-based learning. These findings enhance our understanding of multimodal perception and generative mechanisms in foundation models (fundamental AI models).

- *How I obtained the opportunity*

  This is an independent research project I conceived and executed solo. My research was accepted and presented at the conference.

- *Primary author (if not me)*

  N/A — I am the only author.

- *Scholarly Guidance and Oversight*

  None (independent/self-directed)

- Linguistic Inputs, Visual Outputs: Human Intuition and AI (LLM) Semantic Reasoning in Shape Imagination

Cognitive Systems Research *(an academic journal)*
Manuscript Submitted for Review

- *Abstract*

Humans can intuitively associate phonological forms with visual properties, exemplified by sound symbolism effects such as *"bouba–kiki,"* where people consistently match certain pseudowords with particular shapes. In contrast, AI large language models (LLMs) like GPT-4 learn from text and may rely on learned semantic and distributional regularities rather than perceptual intuition. We compare humans and an LLM in a shape imagination task using non-symbolic stimuli: six novel nonwords (Nonword A– F). Forty multilingual participants (n = 40) read each nonword, imagined its pronunciation, and depicted a simple shape, while GPT-4 received the same strings as text input and generated a single shape description per stimulus. Human responses showed consensus consistent with intuitive sound-symbolic mapping. GPT-4, by contrast, produced shapes that diverged from the human majority, suggesting a reliance on language-based associations rather than pre-linguistic crossmodal intuition. The resulting mismatch highlights a cognitive gap between human intuitive processing and the LLM's semantic reasoning, and provides empirical evidence relevant to the symbol grounding problem. Our findings motivate future work on perceptual grounding and multimodal learning in foundation models.

- Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction *(Spotlight)*

Helmholtz AI Conference 2025

- *Abstract*

The Mallampati score is widely used to assess the difficulty of intubation and endoscopic procedures but suffers from high inter-rater variability. We present an AI model that automatically classifies Mallampati scores and predicts the need for endoscopic assistance using multimodal data. A dataset of 500 patients from a hospital in Korea was used, which includes Mallampati photographs and clinical variables. A Vision Transformer (ViT) processes imaging data, and gradient boosting is applied to clinical features. These AI models are fused to predict Mallampati classes (I–IV) and the binary outcome of endoscopy necessity. Additionally, we integrate a large language model (LLM) to generate clinically interpretable explanations to support decision-making. Our model achieves an accuracy of approximately 92.7% in Mallampati classification and an AUROC of 0.9 for endoscopy necessity prediction. This highlights the potential of deep learning interpretability to standardize airway assessments and support procedural planning.

- *How I obtained the opportunity*
  Secured via direct outreach to Research Professor Haneol Cho (Sejong University); joined as a research collaborator with his support and access to his lab's dataset. My research was accepted for a spotlight presentation which was carried out at the conference.

- *Primary author (if not me)*
  I am the first author and presenter.

- *Scholarly Guidance and Oversight*
  Dr. Haneol Cho – Collaborator, Supporter